SCC

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```
This file is part of CasADi.

CasADi -- A symbolic framework for dynamic optimization.
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```


[1]:
```
import casadi as ca
import numpy
```

Let's construct a block diagonal structure

[2]:
```
A = ca.diagcat(1,ca.DM([[2,3],[3,4]]),ca.DM([[5,6,7],[6,8,9],[7,9,10]]),11)
print(A)
A.sparsity().spy()
```

```
[[1, 00, 00, 00, 00, 00, 00],
 [00, 2, 3, 00, 00, 00, 00],
 [00, 3, 4, 00, 00, 00, 00],
 [00, 00, 00, 5, 6, 7, 00],
 [00, 00, 00, 6, 8, 9, 00],
 [00, 00, 00, 7, 9, 10, 00],
 [00, 00, 00, 00, 00, 00, 11]]
*…
```

```
.**…
.**…
…***.
…***.
…***.
…*
```

[3]:
```
numpy.random.seed(2)
```

We randomly permute this nice structure

[4]:
```
perm =  list(numpy.random.permutation(list(range(A.size1()))))
AP = A[perm,perm]
```

[5]:
```
print(AP)
AP.sparsity().spy()
```

```
[[8, 00, 6, 00, 00, 9, 00],
 [00, 2, 00, 3, 00, 00, 00],
 [6, 00, 5, 00, 00, 7, 00],
 [00, 3, 00, 4, 00, 00, 00],
 [00, 00, 00, 00, 11, 00, 00],
 [9, 00, 7, 00, 00, 10, 00],
 [00, 00, 00, 00, 00, 00, 1]]
*.*..*.
.*.*…
*.*..*.
.*.*…
…*..
*.*..*.
…*
```

And use scc to recover the blocks

[6]:
```
n,p,r = AP.sparsity().scc()
```

[7]:
```
APrestored = AP[p,p]
```

[8]:
```
print(APrestored)
APrestored.sparsity().spy()
print("# blocks: ", n)
print("block boundaries: ", r[:n])
```

```
[[8, 6, 9, 00, 00, 00, 00],
 [6, 5, 7, 00, 00, 00, 00],
 [9, 7, 10, 00, 00, 00, 00],
 [00, 00, 00, 2, 3, 00, 00],
```

```
 [00, 00, 00, 3, 4, 00, 00],
 [00, 00, 00, 00, 00, 11, 00],
 [00, 00, 00, 00, 00, 00, 1]]
***…
***…
***…
…**..
…**..
…*.
…*
# blocks:  4
block boundaries:  [0, 3, 5, 6]
```